

Enriching continuous Lagrange finite element approximation spaces using neural networks

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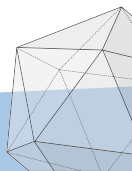
Joint work with:

H. Barucq, F. Faucher, N. Victorion and V. Michel-Dansac.

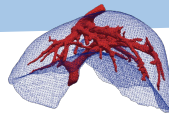
PhD student seminar

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Scientific context

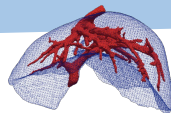


Context: Create real-time digital twins of an organ (e.g. liver).

Objective: Develop an hybrid finite element / neural network method.

accurate quick + parameterized

Scientific context



Context: Create real-time digital twins of an organ (e.g. liver).

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Parametric Poisson problem (with homogeneous Dirichlet BC):

For one or several $\mu \in \mathcal{M}$, find $u : \Omega \rightarrow \mathbb{R}$ such that

$$\begin{cases} -\Delta u(\mathbf{x}; \mu) = f(\mathbf{x}; \mu), & (\mathbf{x}, \mu) \in \Omega \times \mathcal{M}, \\ u(\mathbf{x}; \mu) = 0, & (\mathbf{x}, \mu) \in \partial\Omega \times \mathcal{M}, \end{cases} \quad (\mathcal{P})$$

with $\Omega \subset \mathbb{R}^d$ a domain (d the spatial dimension).

Classical approaches

Physics-Informed Neural Networks

Standard PINNs: Find the optimal weights θ^* , such that

$$\theta^* = \underset{\theta}{\operatorname{argmin}} \left(\omega_r J_r(\theta) + \omega_b J_b(\theta) \right), \quad (\mathcal{P}_\theta)$$

with

residual loss

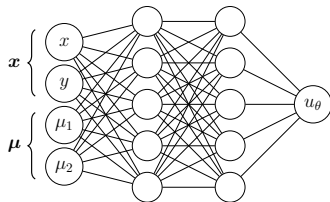
$$J_r(\theta) = \int_{\mathcal{M}} \int_{\Omega} \left| \Delta u_\theta(\mathbf{x}, \boldsymbol{\mu}) + f(\mathbf{x}, \boldsymbol{\mu}) \right|^2 d\mathbf{x} d\boldsymbol{\mu},$$

boundary loss

$$J_b(\theta) = \int_{\mathcal{M}} \int_{\partial\Omega} \left| u_\theta(\mathbf{x}, \boldsymbol{\mu}) \right|^2 d\mathbf{x} d\boldsymbol{\mu},$$

with ω_r and ω_b are some weights.

u_θ is a neural network, e.g. fully-connected NN.
(see example in 2D with 2 parameters)



Monte-Carlo method: Discretize the cost functions by random process.

Finite Element Method¹

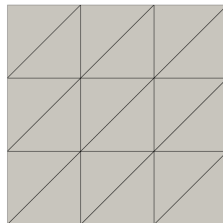
Variational Problem:

$$\text{Find } u_h \in V_h^0 \text{ such that, } \forall v_h \in V_h^0, a(u_h, v_h) = l(v_h), \quad (\mathcal{P}_h)$$

with h the characteristic mesh size, a and l the bilinear and linear forms given by

$$a(u_h, v_h) = \int_{\Omega} \nabla u_h \cdot \nabla v_h, \quad l(v_h) = \int_{\Omega} f v_h,$$

and V_h^0 a continuous, piecewise polynomial space of degree k .



Cartesian mesh

¹[Ern and Guermond, 2004]

Finite Element Method¹

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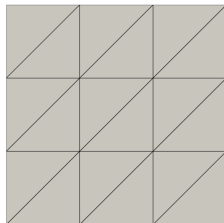
Linear system: Let $(\phi_1, \dots, \phi_{N_{\text{dofs}}})$ a basis of V_h^0 .

Find $U \in \mathbb{R}^{N_{\text{dofs}}}$ such that

$$AU = b,$$

$$A = (a(\phi_i, \phi_j))_{1 \leq i, j \leq N_{\text{dofs}}} \in \mathbb{R}^{N_{\text{dofs}} \times N_{\text{dofs}}},$$

$$b = (l(\phi_j))_{1 \leq j \leq N_{\text{dofs}}} \in \mathbb{R}^{N_{\text{dofs}}}.$$



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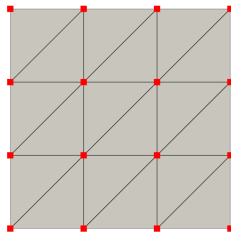
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Degrees of freedom ($k = 1$)

¹[Ern and Guermond, 2004]

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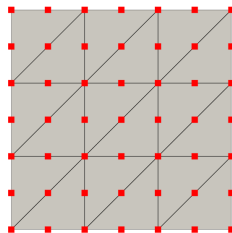
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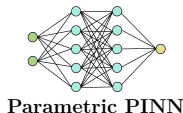
Enriched finite element method using PINNs

Pipeline of the Enriched FEM

OFFLINE : PINN training

Inputs

$\mathbf{x} \in \Omega$: space coordinates
 $\boldsymbol{\mu} \in \mathcal{M}$: parameter



Output

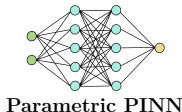
$u_{\theta}(\mathbf{x}, \boldsymbol{\mu})$: prediction of
 the PDE solution $u(\mathbf{x}, \boldsymbol{\mu})$

Pipeline of the Enriched FEM

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$u_{\theta}(\mathbf{x}, \boldsymbol{\mu})$: prediction of
 the PDE solution $u(\mathbf{x}, \boldsymbol{\mu})$

ONLINE (fixed $\boldsymbol{\mu}$) : PINN evaluation + Enriched FEM resolution

PINN evaluation

$u_{\theta}(\cdot, \boldsymbol{\mu})$: prediction for
 the given parameter $\boldsymbol{\mu}$



FEM solver
 enriched by u_{θ}



Output

enriched FEM solution
 (depending on the mesh size h)

Complete ONLINE process : quick + accurate

Additive approach

Variational Problem: Let $u_\theta \in H^{k+1}(\Omega) \cap H_0^1(\Omega)$ be a PINN prior.

$$\text{Find } c_h^+ \in V_h^0 \text{ such that, } \forall v_h \in V_h^0, a(c_h^+, v_h) = l(v_h) - a(u_\theta, v_h), \quad (\mathcal{P}_h^+)$$

with the **enriched trial space** V_h^+ defined by

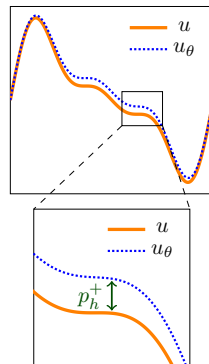
$$V_h^+ = \{u_h^+ = u_\theta + c_h^+, \quad c_h^+ \in V_h^0\}.$$

General Dirichlet BC: If $u = g$ on $\partial\Omega$, then

$$c_h^+ = g - u_\theta \quad \text{on } \partial\Omega,$$

with u_θ the PINN prior.

We expect that the modified problem will give the same results as the standard one on coarser meshes.



Convergence analysis

Theorem 1: Convergence analysis of the standard FEM [Ern and Guermond, 2004]

We denote $u_h \in V_h$ the solution of $(\mathcal{P}_h)$ with V_h the standard trial space. Then,

$$\|u - u_h\|_{L^2} \leqslant C_{L^2} h^{k+1} |u|_{H^{k+1}}.$$

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Theorem 2: Convergence analysis of the enriched FEM [Lecourtier et al., 2025]

We denote $u_h^+ \in V_h^+$ the solution of $(\mathcal{P}_h^+)$ with V_h^+ the enriched trial space. Then,

$$\|u - u_h^+\|_{L^2} \leqslant \frac{|u - u_\theta|_{H^{k+1}}}{|u|_{H^{k+1}}} (C_{L^2} h^{k+1} |u|_{H^{k+1}}).$$

$$\frac{|u - u_\theta|_{H^{k+1}}}{|u|_{H^{k+1}}} < 1 \Rightarrow \text{Gains of the additive approach.}$$

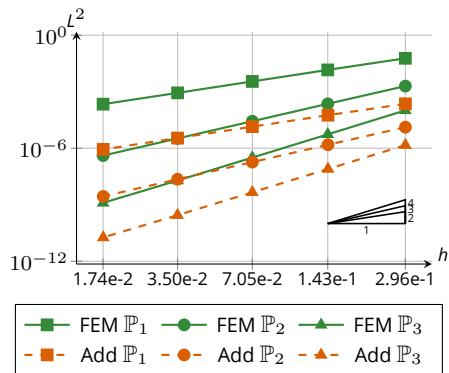
Numerical results

2D Poisson problem on Square - Dirichlet BC

Online phase - 1 set of parameters

$$\mu^{(1)} = (0.05, 0.22)$$

Error estimates in L^2 norm:



LECOURTIER Frédérique

Parametric framework

Considering the set of $n_p = 50$ parameter instances

$$\mathcal{S} = \left\{ \boldsymbol{\mu}^{(1)}, \dots, \boldsymbol{\mu}^{(n_p)} \right\}.$$

Gains achieved: $\|u - u_h\|_{L^2} / \|u - u_h^+\|_{L^2}$

Gains in L^2 rel error of our method w.r.t. FEM			
k	min	max	mean
1	134.32	377.36	269.39
2	67.02	164.65	134.85
3	39.52	72.65	61.55

Cartesian mesh : N^2 nodes with $N = 20$.

Numerical results

3D Poisson problem on Cube - Dirichlet BC

Problem considered

Problem statement: Consider the Poisson problem in 3D with Dirichlet BC:

$$\begin{cases} -\Delta u = f, & \text{in } \Omega \times \mathcal{M}, \\ u = 0, & \text{on } \partial\Omega \times \mathcal{M}, \end{cases}$$

with $\Omega = [-0.5\pi, 0.5\pi]^3$ and $\mathcal{M} = [-0.5, 0.5]^3$ ($p = 3$ parameters).

Analytical solution :

$$u(\mathbf{x}, \mu) = \exp\left(-\frac{(x - \mu_1)^2 + (y - \mu_2)^2 + (z - \mu_3)^2}{2}\right) \sin(2x) \sin(2y) \sin(2z).$$

PINN training:

MLP 5 layers (tanh - 40, 60, 60, 60, 40); LBFGs optimizer (5000 epochs - 40000 col. pts).

Imposing the Dirichlet BC exactly in the PINN with the levelset φ defined by

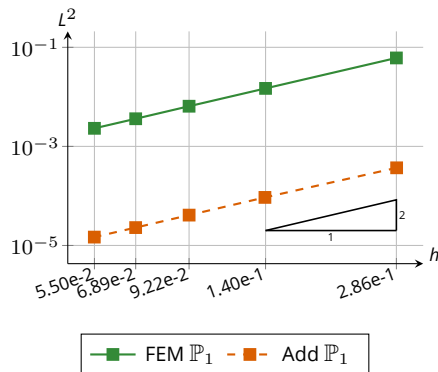
$$\varphi(\mathbf{x}) = (x + 0.5\pi)(x - 0.5\pi)(y + 0.5\pi)(y - 0.5\pi)(z + 0.5\pi)(z - 0.5\pi).$$

Training time : 12 minutes on NVIDIA RTX 2000 (8GB VRAM).

Online phase - 1 parameters instance I

$$\mu^{(1)} = (0.05, 0.22, 0.1)$$

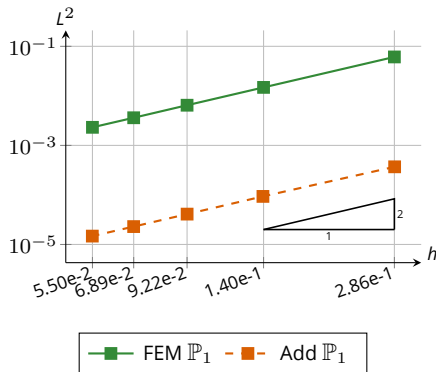
Error estimates in L^2 norm:



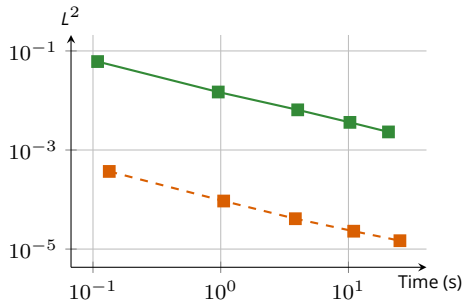
Online phase - 1 parameters instance I

$$\mu^{(1)} = (0.05, 0.22, 0.1)$$

Error estimates in L^2 norm:



L^2 error w.r.t. execution time*:

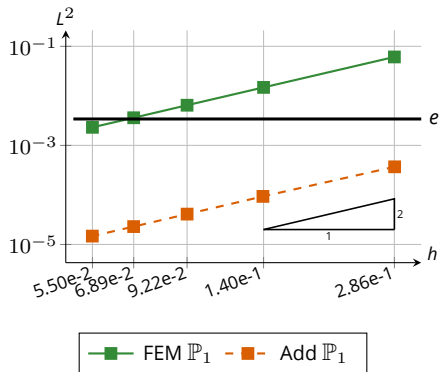


* mesh + assembly + solve
 + prior derivatives evaluation
 + training time

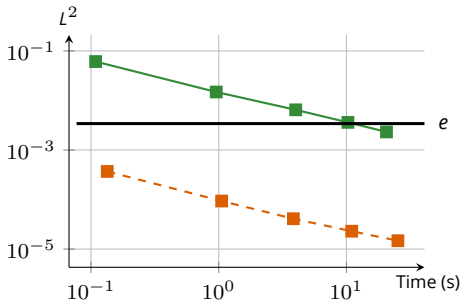
Online phase - 1 parameters instance I

$$\mu^{(1)} = (0.05, 0.22, 0.1)$$

Error estimates in L^2 norm:



L^2 error w.r.t. execution time*:



- * mesh + assembly + solve
- + prior derivatives evaluation
- + training time

Online phase - 1 parameters instance II

$$\mu^{(1)} = (0.05, 0.22, 0.1)$$

N_{dofs} and computation time required to reach the same error e :

e	N		N_{dofs}		Computation time*	
	FEM	Add	FEM	Add	FEM	Add
$1 \cdot 10^{-3}$	152	12	$3.51 \cdot 10^6$	$1.73 \cdot 10^3$	$(7.65 \cdot 10^1)$	$5.2 \cdot 10^{-2}$
$1 \cdot 10^{-4}$	484	39	$1.13 \cdot 10^8$	$5.93 \cdot 10^4$	$(2.8 \cdot 10^3)$	$9.26 \cdot 10^{-1}$
$1 \cdot 10^{-5}$	1539	122	$3.65 \cdot 10^9$	$1.82 \cdot 10^6$	$(1.02 \cdot 10^5)$	$(5.26 \cdot 10^1)$

— $\div 1471$ —→

Cartesian mesh : N^3 nodes; $\mathbb{P}_1$ elements.

* The results in brackets are extrapolations.

Numerical results

2D Poisson problem on Square - Dirichlet BC

3D Poisson problem on Cube - Dirichlet BC

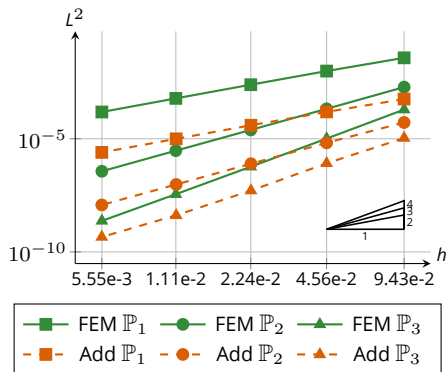
2D Anisotropic Elliptic problem on Square - Dirichlet BC

2D Poisson problem on Annulus - Mixed BC

Online phase

Error estimates: 1 parameters instance.

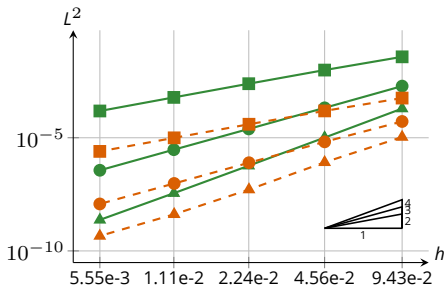
$$\mu^{(1)} = (0.51, 0.54, 0.52, 0.55)$$



Online phase

Error estimates: 1 parameters in-stance.

$$\mu^{(1)} = (0.51, 0.54, 0.52, 0.55)$$



Gains achieved: $\|u - u_h\|_{L^2} / \|u - u_h^+\|_{L^2}$

Considering $n_p = 50$ parameter instances

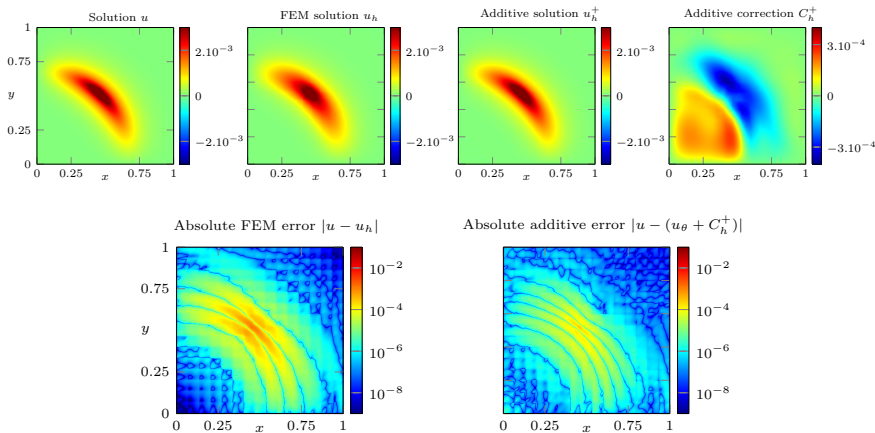
$$\mathcal{S} = \{\mu^{(1)}, \dots, \mu^{(n_p)}\}.$$

**Gains in L^2 rel error
of our method w.r.t. FEM**

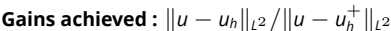
k	min	max	mean
1	7.12	82.57	35.67
2	3.54	35.88	18.32
3	1.33	26.51	8.32

Cartesian mesh : N^2 nodes with $N = 20$.

Numerical solutions and errors

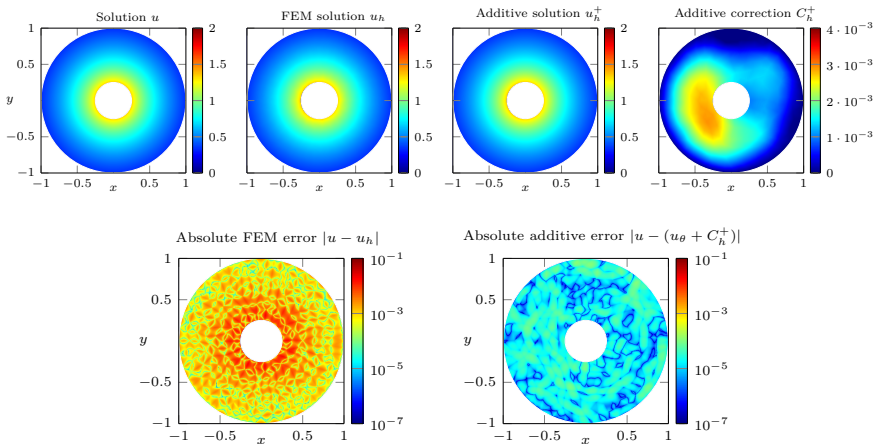


$$\mu^{(2)} = (0.46, 0.52, 0.05, 0.12) \quad (k = 2, N = 16)$$

$$\mu^{(1)} = 2.51$$

$$\mathcal{S} = \left\{ \boldsymbol{\mu}^{(1)}, \dots, \boldsymbol{\mu}^{(n_p)} \right\}.$$

Mesh size: $h = 1.33 \cdot 10^{-1}$

Numerical solutions and errors



$$\mu^{(1)} = 2.51 \quad (k = 1, h = 1.67 \cdot 10^{-1})$$

Extension to non-linear problems

Physics-Informed Neural Network (PINN)

Finite Element Method (FEM)

Enriched finite element method using PINNs

Heated cavity test case¹

Stationary incompressible Navier-Stokes equations (with buoyancy and gravity):

We consider $\mathbf{x} = (x, y) \in \Omega$ and $\mathbf{e}_y = (0, 1)$.

Find $\mathbf{U} = (\mathbf{u}, p, T) = (u_1, u_2, p, T)$ such that

$$\begin{cases} R_{mom}(\mathbf{U}; \mathbf{x}, \boldsymbol{\mu}) = 0 & \text{in } \Omega & \text{(momentum)} \\ R_{inc}(\mathbf{U}; \mathbf{x}, \boldsymbol{\mu}) = 0 & \text{in } \Omega & \text{(incompressibility)} \\ R_{ener}(\mathbf{U}; \mathbf{x}, \boldsymbol{\mu}) = 0 & \text{in } \Omega & \text{(energy)} \end{cases} \quad (\mathcal{P})$$

with $g = 9.81$ the gravity, $\beta = 0.1$ the expansion coefficient, $\nu = 0.01$ the viscosity and $k_f = 0.01$ the thermal conductivity. (Rayleigh number: 156 960)

Boundary Conditions:

No-slip BC : $\mathbf{u} = 0$ on $\partial\Omega$ **Isothermal BC :** $T = 1$ on the left wall ($x = -1$)
 $T = -1$ on the right wall ($x = 1$)

Adiabatic BC : $\frac{\partial T}{\partial n} = 0$ on the top and bottom walls ($y = \pm 1$, denoted by Γ_{ad})

¹Non-parametric study.

Extension to non-linear problems

Physics-Informed Neural Network (PINN)

Finite Element Method (FEM)

PINN training

Approximate the solution of ($\mathcal{P}$) by a PINN : Find the optimal weights θ^* , such that

$$\theta^* = \underset{\theta}{\operatorname{argmin}} \left(J_{inc}(\theta) + J_{mom}(\theta) + J_{ener}(\theta) + J_{ad}(\theta) \right), \quad (\mathcal{P}_\theta)$$

where the different cost functions¹ are defined by

adiabatic condition

$$J_{ad}(\theta) = \int_{\mathcal{M}} \int_{\Gamma_{ad}} \left| \frac{\partial T_\theta(\mathbf{x}, \mu)}{\partial n} \right|^2 d\mathbf{x} d\mu,$$

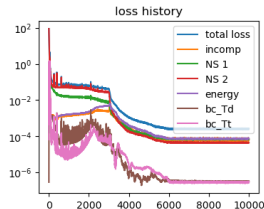
3 residual losses

$$J_\bullet(\theta) = \int_{\mathcal{M}} \int_{\Omega} |R_\bullet(U_\theta(\mathbf{x}, \mu); \mathbf{x}, \mu)|^2 d\mathbf{x} d\mu,$$

with U_θ the non-parametric NN and $\bullet$ the PDE considered (i.e. *inc*, *mom* or *ener*).

Network - MLP		Training (ADAM / LBFGs)			
<i>layers</i>	40, 60, 60, 60, 40	<i>lr</i>	7e-3	N_{col}	40000
σ	sine	<i>nepochs</i>	10000	N_{bc}	30000

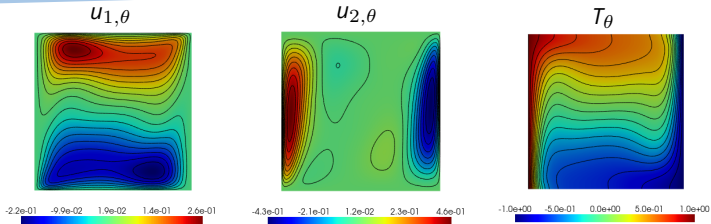
+ some weights to balance the different losses.



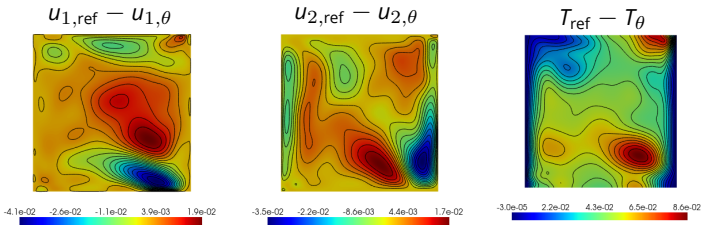
¹ Discretized by a random process using Monte-Carlo method.

Non parametric PINN prediction

Prediction:



Error map¹:



L^2 error:
(relative)

7.60×10^{-2}

5.38×10^{-2}

9.63×10^{-2}

¹Reference solution computed by FEM on an over-refined mesh.

Discrete variational formulation

We consider a mixed finite element space $M_h = [v_h^0]^2 \times Q_h \times W_h$ (4 unknowns).

Variational Problem: Find $U_h = (\mathbf{u}_h, p_h, T_h) \in M_h$ s.t., $\forall (\mathbf{v}_h, q_h, w_h) \in M_h^0$,

$$\begin{aligned}
 & \int_{\Omega} (\mathbf{u}_h \cdot \nabla) \mathbf{u}_h \cdot \mathbf{v}_h \, d\mathbf{x} + \mu \int_{\Omega} \nabla \mathbf{u}_h : \nabla \mathbf{v}_h \, d\mathbf{x} \\
 & \quad - \int_{\Omega} p_h \nabla \cdot \mathbf{v}_h \, d\mathbf{x} - g \int_{\Omega} (1 + \beta T_h) \mathbf{e}_y \cdot \mathbf{v}_h \, d\mathbf{x} = 0, \quad (\text{momentum}) \\
 & \int_{\Omega} q_h \nabla \cdot \mathbf{u}_h \, d\mathbf{x} + 10^{-4} \int_{\Omega} q_h p_h \, d\mathbf{x} = 0, \quad (\text{incompressibility + pressure penalization}) \\
 & \int_{\Omega} (\mathbf{u}_h \cdot \nabla T_h) w_h \, d\mathbf{x} + \int_{\Omega} k_f \nabla T_h \cdot \nabla w_h \, d\mathbf{x} = 0, \quad (\text{energy})
 \end{aligned} \tag{\mathcal{P}_h}$$

Discrete variational formulation

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 & \quad - \int_{\Omega} p_h \nabla \cdot \mathbf{v}_h \, d\mathbf{x} - g \int_{\Omega} (1 + \beta T_h) \mathbf{e}_y \cdot \mathbf{v}_h \, d\mathbf{x} = 0, \quad (\text{momentum}) \\
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 & \int_{\Omega} (\mathbf{u}_h \cdot \nabla T_h) w_h \, d\mathbf{x} + \int_{\Omega} k_f \nabla T_h \cdot \nabla w_h \, d\mathbf{x} = 0, \quad (\text{energy})
 \end{aligned} \tag{P_h}$$

Non-linear terms $\Rightarrow$ Non-linear system to solve.

Newton method

Denoting N_h the dimension of M_h , we want to solve the non linear system:

$$F(\vec{U}) = 0$$

with $F : \mathbb{R}^{N_h} \rightarrow \mathbb{R}^{N_h}$ the non-linear operator associated to $(\mathcal{P}_h)$ and $\vec{U} \in \mathbb{R}^{N_h}$ the unknown vector.

Algorithm 1: Newton algorithm

Initialization step: set $\vec{U}^{(0)} = \vec{U}_0$;

for $n \geq 0$ **do**

 Solve the linear system $F(\vec{U}^{(n)}) + F'(\vec{U}^{(n)})\delta^{(n+1)} = 0$ for $\delta^{(n+1)}$;

 Update $\vec{U}^{(n+1)} = \vec{U}^{(n)} + \delta^{(n+1)}$;

end

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⚠ The choice of the $\vec{U}_0$ is crucial for the convergence of the Newton method.

Solution: Continuation method where we progressively increase the non-linearity of the problem.

Extension to non-linear problems

Finite Element Method (FEM)

Enriched finite element method using PINNs

Additive approach

Considering the PINN prior $U_\theta = (\mathbf{u}_\theta, p_\theta, T_\theta)$, we define the **mixed finite element space additively enriched** by the PINN as follows:

$$M_h^+ = \{U_h^+ = U_\theta + C_h^+, \quad C_h^+ \in M_h^0\}$$

with $M_h^0 = [V_h^0]^2 \times Q_h \times W_h^0$, $U_h^+ = (\mathbf{u}_h^+, p_h^+, T_h^+) \in M_h^+$ and $C_h^+ = (\mathbf{C}_{h,u}^+, C_{h,p}^+, C_{h,T}^+)$.

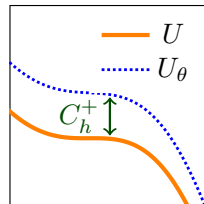
We can then define the three finite element subspaces of M_h^+ as follows:

$$V_h^+ = \{\mathbf{u}_h^+ = \mathbf{u}_\theta + \mathbf{C}_{h,u}^+, \quad \mathbf{C}_{h,u}^+ \in [V_h^0]^2\},$$

$$Q_h^+ = \{p_h^+ = p_\theta + C_{h,p}^+, \quad C_{h,p}^+ \in Q_h\},$$

$$W_h^+ = \{T_h^+ = T_\theta + C_{h,T}^+, \quad C_{h,T}^+ \in W_h^0\},$$

where $\mathbf{C}_{h,u}^+$, $C_{h,p}^+$ and $C_{h,T}^+$ becomes the unknowns.



Modified variational formulation

Weak problem : Find $C_h^+ = (\mathbf{c}_{h,u}^+, \mathbf{c}_{h,p}^+, C_{h,T}^+) \in M_h^0$ s.t., $\forall (\mathbf{v}_h, q_h, w_h) \in M_h^0$,

$$\begin{aligned} & \int_{\Omega} [(\mathbf{u}_{\theta} \cdot \nabla) \mathbf{u}_{\theta} + (\mathbf{u}_{\theta} \cdot \nabla) \mathbf{c}_{h,u}^+ + (\mathbf{c}_{h,u}^+ \cdot \nabla) \mathbf{u}_{\theta} + (\mathbf{c}_{h,u}^+ \cdot \nabla) \mathbf{c}_{h,u}^+] \cdot \mathbf{v}_h \, d\mathbf{x} \\ & + \mu \left(\int_{\Omega} \nabla \mathbf{u}_{\theta} : \nabla \mathbf{v}_h \, d\mathbf{x} + \int_{\Omega} \nabla \mathbf{c}_{h,u}^+ : \nabla \mathbf{v}_h \, d\mathbf{x} \right) + \left(\int_{\Omega} \nabla p_{\theta} \cdot \mathbf{v}_h \, d\mathbf{x} - \int_{\Omega} \mathbf{c}_{h,p}^+ \nabla \cdot \mathbf{v}_h \, d\mathbf{x} \right) \\ & - g \int_{\Omega} (1 + \beta(T_{\theta} + C_{h,T}^+)) \mathbf{e}_y \cdot \mathbf{v}_h \, d\mathbf{x} = 0, \text{ (momentum)} \\ & \int_{\Omega} q_h [\nabla \cdot \mathbf{u}_{\theta} + \nabla \cdot \mathbf{c}_{h,u}^+] \, d\mathbf{x} + 10^{-4} \int_{\Omega} q_h (p_{\theta} + \mathbf{c}_{h,p}^+) \, d\mathbf{x} = 0, \text{ (incompressibility + penal)} \\ & \int_{\Omega} [\mathbf{u}_{\theta} \cdot \nabla T_{\theta} + \mathbf{u}_{\theta} \cdot \nabla C_{h,T}^+ + \mathbf{c}_{h,u}^+ \cdot \nabla T_{\theta} + \mathbf{c}_{h,u}^+ \cdot \nabla C_{h,T}^+] w_h \, d\mathbf{x} \\ & + k_f \left(\int_{\Omega} \nabla T_{\theta} \cdot \nabla w_h \, d\mathbf{x} + \int_{\Omega} \nabla C_{h,T}^+ \cdot \nabla w_h \, d\mathbf{x} \right) = 0, \text{ (energy)} \end{aligned} \quad (\mathcal{P}_h^+)$$

with $U_{\theta} = (\mathbf{u}_{\theta}, p_{\theta}, T_{\theta})$ the PINN prior and some modified boundary conditions.

Newton method - Additive approach

Denoting N_h the dimension of M_h^+ , we want to solve the non-linear system:

$$F_\theta(\vec{C}) = 0$$

with $F_\theta : \mathbb{R}^{N_h} \rightarrow \mathbb{R}^{N_h}$ the non linear operator associated to the weak problem ($\mathcal{P}_h^+$) and $\vec{C} \in \mathbb{R}^{N_h}$ the **correction vector (unknown)**.

Algorithm 2: Newton algorithm

Initialization step: set $\vec{C}^{(0)} = \mathbf{0}$;

for $n \geq 0$ **do**

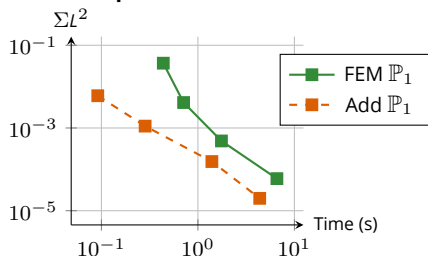
 Solve the linear system $F_\theta(\vec{C}^{(n)}) + F'_\theta(\vec{C}^{(n)})\delta^{(n+1)} = 0$ for $\delta^{(n+1)}$;

 Update $\vec{C}^{(n+1)} = \vec{C}^{(n)} + \delta^{(n+1)}$;

end

Numerical results

N_{dofs} and computation time required to reach the same error e :



e	N_{dofs}		Online time	
	FEM	Add	FEM	Add
$1 \cdot 10^{-3}$	33 204	13 764	1.29	0.31
$1 \cdot 10^{-4}$	150 339	70 303	4.76	1.78
$1 \cdot 10^{-5}$	690 924	339 231	20.34	6.42

Conclusion

TODO

Preprint (linear)



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Preprint (linear)

